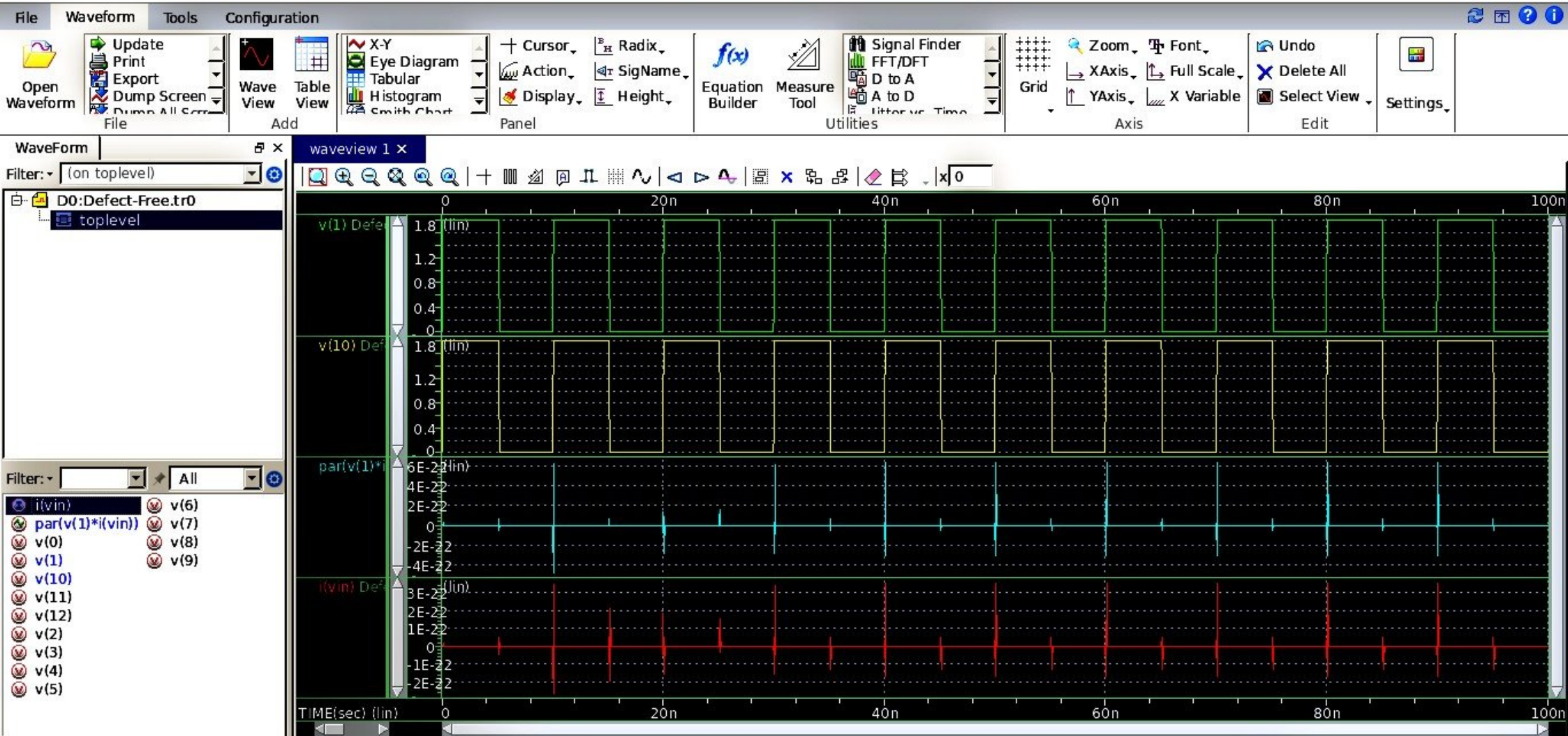
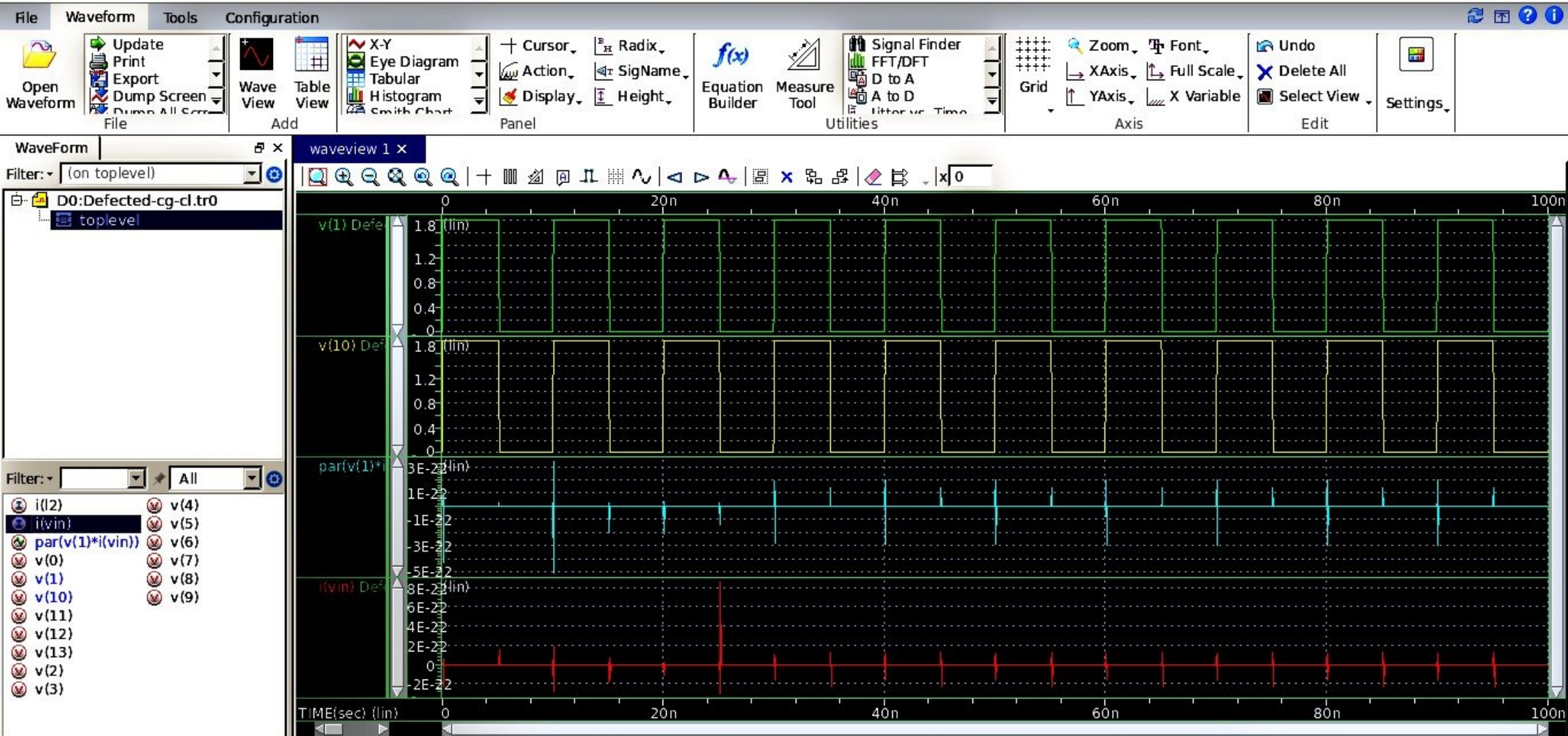


Custom WaveView Version N-2017.12-SP2 Synopsys, Inc.



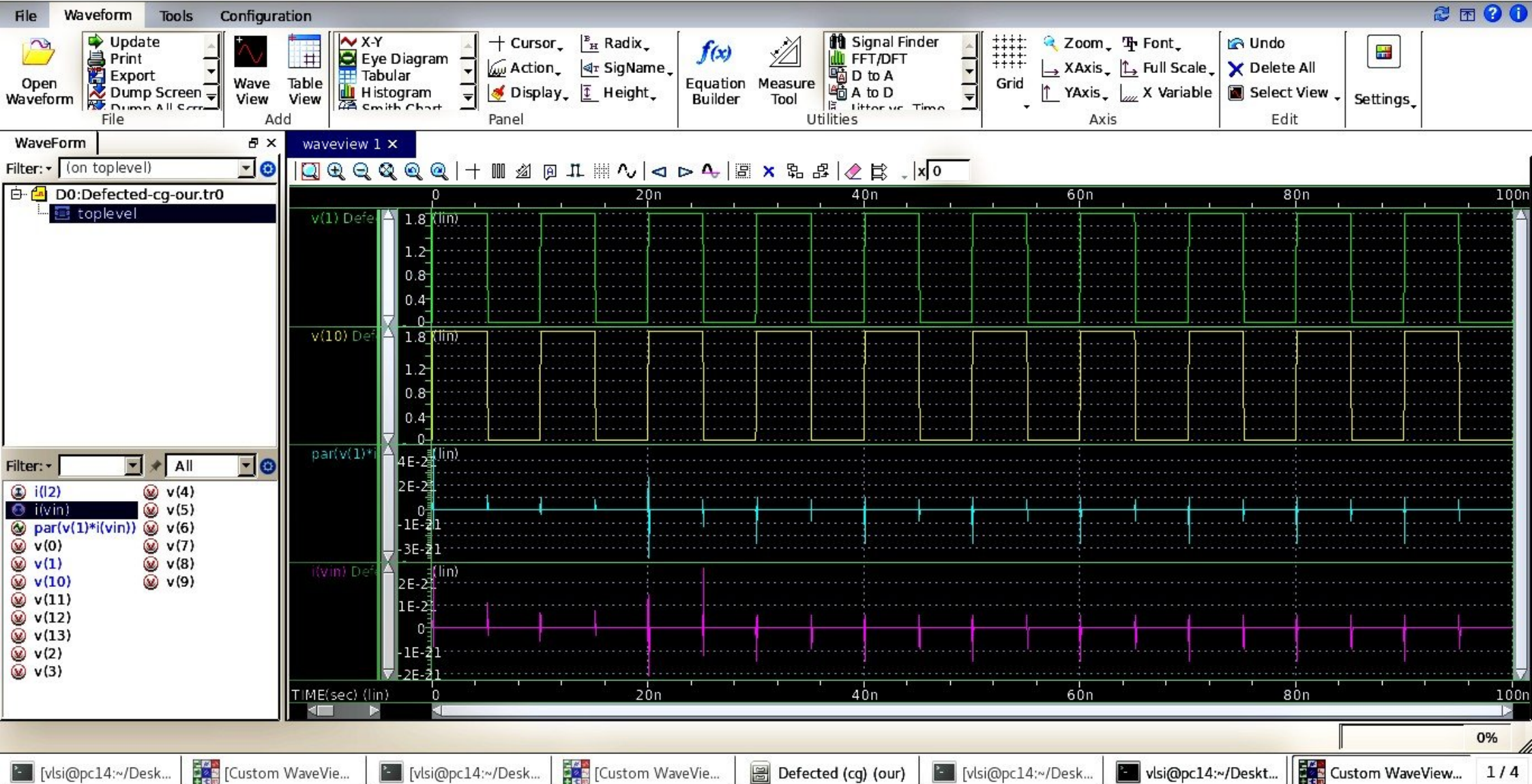
0%

Custom WaveView Version N-2017.12-SP2 Synopsys, Inc.



0%

Custom WaveView Version N-2017.12-SP2 Synopsys, Inc.



```
$DATA1 SOURCE='HSPICE' VERSION='0-2018.09-2 linux64' PARAM_COUNT=0
.TITLE '.temp25'
tr          tf          pavg          temper
alter#
-5.394e-23  1.252e-20    4.149e-25    25.0000
1
```

Open

Defected-cg-cl.mt0

~/Desktop/OIP-k.d.s.k/Defected (cg) (chaoliu)

Save

≡

-

□

✕

```
$DATA1 SOURCE='HSPICE' VERSION='0-2018.09-2 linux64' PARAM_COUNT=0
```

```
.TITLE '.temp25'
```

```
tr          tf          pavg          temper
alter#
3.981e-23   -3.085e-22    1.931e-25    25.0000
```

```
1
```

Loading file '/home/vlsi/Desktop/OIP-k.d.s.k/Defected (cg) (chaoliu)/Defected-cg-cl.mt0'...

Plain Text ▾ Tab Width: 8 ▾

Ln 1, Col 1 ▾

INS

[vlsi@pc14:~/Desk...

[Custom WaveVie...

[vlsi@pc14:~/Desk...

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[vlsi@pc14:~/Desk...

[Custom WaveVie...

Defected (cg) (cha...

Defected-cg-cl.mt...

1 / 4

Open

Defected-cg-our.mt0

~/Desktop/OIP-k.d.s.k/Defected (cg) (our)

Save

-

□

✕

```
$DATA1 SOURCE='HSPICE' VERSION='0-2018.09-2 linux64' PARAM_COUNT=0
```

```
.TITLE '.temp25'
```

```
tr          tf          pavg          temper
alter#
3.127e-23    1.422e-20    -5.465e-24    25.0000
```

```
1
```

Loading file '/home/vlsi/Desktop/OIP-k.d.s.k/Defected (cg) (our)/Defected-cg-our.mt0'...

Plain Text ▾ Tab Width: 8 ▾

Ln 1, Col 1 ▾

INS

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[vlsi@pc14:~/Desk...

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[vlsi@pc14:~/Desk...

[Custom WaveVie...

Defected (cg) (our)

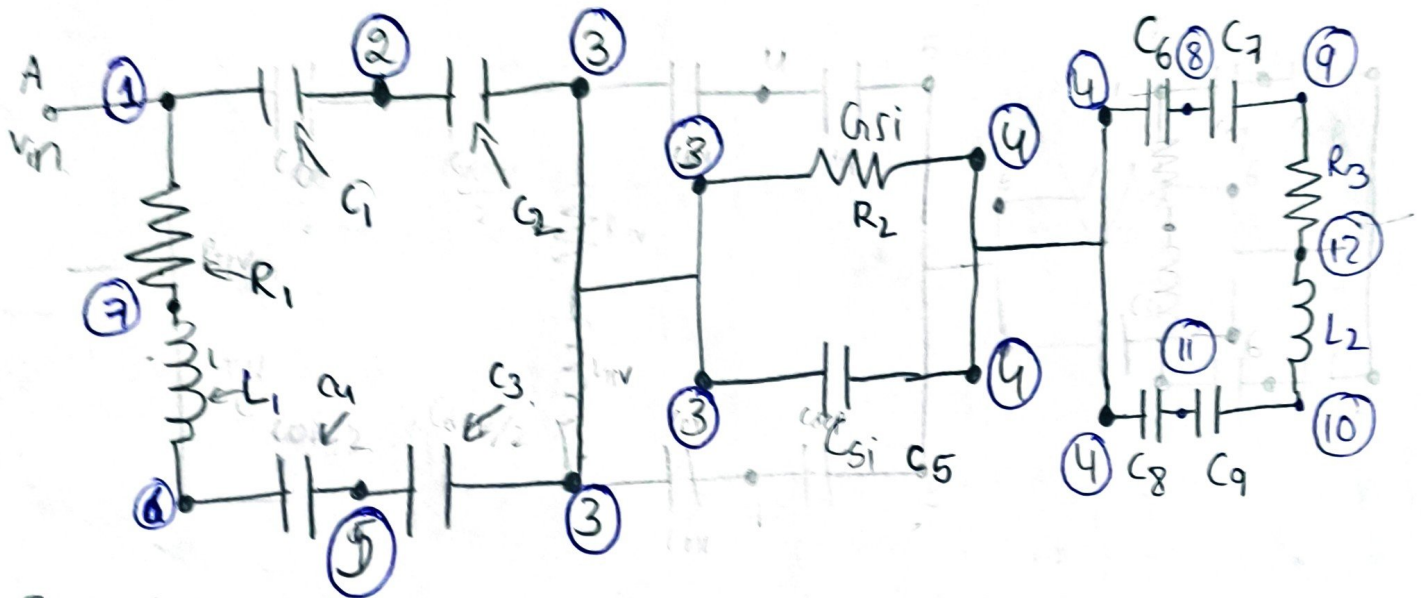
Defected-cg-our.m...

1 / 4

22/06/26 Day 17 Kamineni Dhana Sri Keelthi

Hspice (Noise, crosstalk, delay)

Defect-free Simulation



From matlab

$$R_{TSV} = 8.284 \times 10^{-2} \Omega$$

$$L_{TSV} = 2.097 \times 10^{-12} H$$

$$G_{Si} = 3.540 \times 10^{-4} S$$

$$C_{Si} = 3.730 \times 10^{-15} F$$

$$C_{ox} = 6.50714 \times 10^{-14} F$$

$$C_{dep} = 3.584 \times 10^{-14} F$$

$$82.8 \text{ m}\Omega$$

$$2.09 \text{ pH}$$

$$854 \text{ }\mu S$$

$$3.73 \text{ fF}$$

$$65.07 \text{ fF}$$

$$35.84 \text{ fF}$$

Code

• temp25

*** Supply voltage ***

Vin 0 pulse [0 1.8 0s 100p 100p 5n 10n]

*** Network ***

R1	1	7	82m
R2	3	4	354u
L1	7	6	2.0p
C1	1	2	32.53f
C2	2	3	17.92f
C3	5	3	17.92f
C4	6	5	32.53f
C5	3	4	3.73f

R3	9	12	82m
L2	12	10	2p
C6	4	8	17.92f
C7	8	9	32.53f
C8	4	11	17.92f
C9	11	12	32.53f

* operation part *

• Option post

• trans

*** measure ***

next
delay
close all.

later

now rise & fall time.

• measure tran tr trig v(1) val=0.9 rise=1 targ v(10)

+ val=0.9 rise=1

• measure tran tf trig v(1) val=0.9 fall=1 targ v(10)

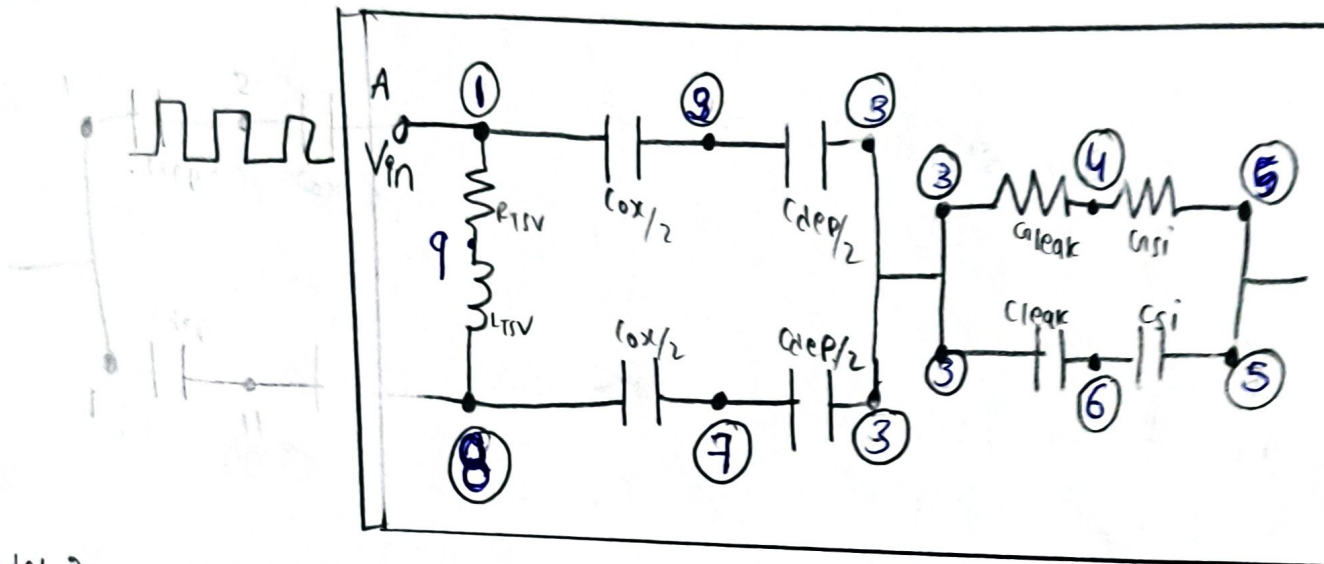
+ val=0.9 fall=1

100126 Day 17 kamineni Dhana Sri keerthi

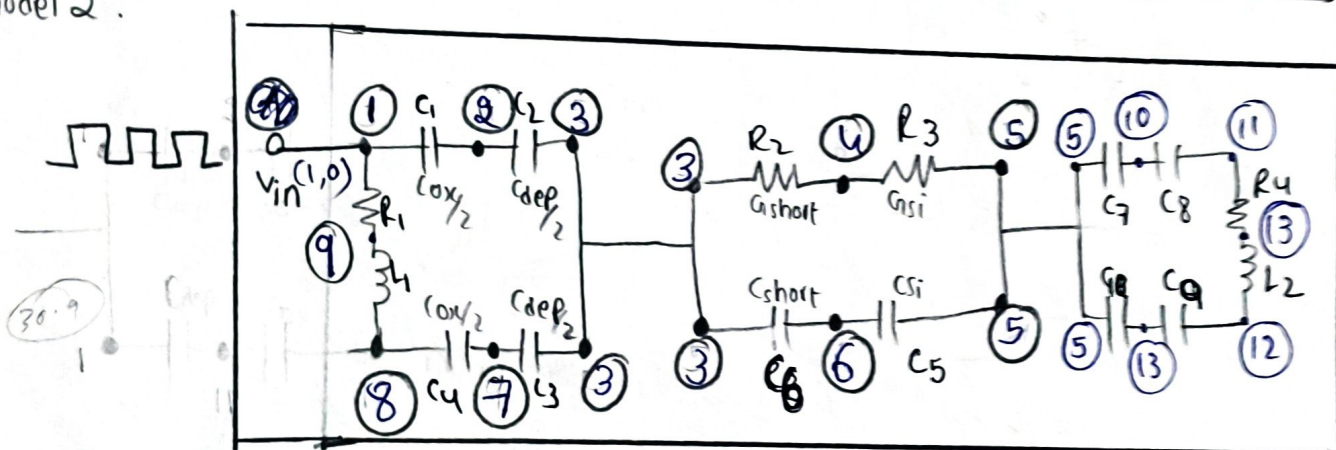
Hspice (Noise, crosstalk, delay)

Defected CG (2models (chao liu; our))

chao liu.



model 2.



Values (chao liu)

From matlab.

$$R_{tsv} = 82.8 \text{ m}\Omega$$

$$L_{tsv} = 2.09 \text{ PH}$$

$$C_{ox} = 65.0 \text{ FF}$$

$$C_{dep} = 34.4 \text{ FF}$$

$$C_{si} = 2.73 \text{ FF}$$

$$G_{si} = 259.8 \text{ }\mu\text{S}$$

$$C_{leak} = 64 \text{ FF}$$

$$G_{leak} = -214 \text{ }\mu\text{S}$$

$$82.8 \text{ m}\Omega$$

$$2.09 \text{ PH}$$

$$C_{ox_new} = 61.8 \text{ FF}$$

Proposed

$$R_{tsv} =$$

$$L_{tsv} =$$

$$C_{ox_new} =$$

$$C_{dep} =$$

$$C_{si} =$$

$$G_{si} =$$

$$C_{short} = 14.09 \text{ FF}$$

$$G_{short} = 108.4 \text{ }\mu\text{S}$$

same

• lemp25

supply

V_{in} 1 0 pulse [0 1.8 0.5 100p 100p 5n 10n]

* Network *

	R_1	1	9	82m	src	C_5	6	5	2.73f
real	R_2	3	4	214 μ	load/short	C_6	3	6	64f
	R_3	4	5	259 μ	dep	C_7	5	10	1790f
	R_4	11	13	82m	ox	C_8	10	11	30.9f
	L_1	9	8	2p	ox	C_9	13	12	30.9f
	L_2	13	12	2p	dep	C_{10}	5	13	17.90f
ox	C_1	1	2	30.9f					
dep	C_2	2	3	17.2f					
dep	C_3	7	3	17.2f					
ox	C_4	8	7	30.9f					

ox $\rightarrow$ our proposed

R_2	3	4	1084 μ
C_6	3	6	14f

• measure tran tr trng v (1) val=0.9 rise=1 larg v(10) val=0.9
rise=1

• - - - -
- - - - -

10/06/26 DAY 17 "Kamineni Dhana Sri Keerthi"

Simulation [Defect free, C_0 , C_L , C_{hour} , along with eq], RL eq derive.

Lsh:-
$$\frac{\mu_0 \mu_{TSV} t_{ins}}{2\pi} \left[\cosh^{-1} \left[\frac{t_{ins}}{\sqrt{L_s^2 + t_{ins}^2}} \right] - \sqrt{2 t_{ins}^2 + L_s^2} + \frac{t_{ins}}{a} + \sqrt{L_s^2 + t_{ins}^2} \right]$$

